Report

Date of seminar : 6-3-2026

Speaker : Dr Ye Jun

Title of the presentation - vQPU: Building Foundational Software Components for Hybrid Interfaces in Hybrid Quantum-Classical Computing

The seminar delivered by Dr. Ye Jun examines the growing bottleneck at the interface between high-level software abstractions and low-level hardware execution. As the field transitions toward hybrid quantum-classical computing (HQCC), engineering limitations have progressed beyond the traditional issues of qubit quality and noise. The new strategy now is to improve integration across the software–hardware stack. The work presented in this seminar addresses this gap by proposing a unified, framework-agnostic infrastructure that connects quantum algorithms to real device behavior. The central idea is that moving from high level/ gate-level abstractions and focussing more instead towards pulse-level and system-level modeling enables more accurate algorithm validation, improved scalability, and tighter hardware–software co-design. Two integrated components namely the EmuPlat platform and the QuanTempo simulation engine are proposed candidates for this job.

EmuPlat is introduced as a full-stack quantum emulation platform capable of operating across different heterogeneous ecosystems such as Qiskit and CUDA-Q. Its core architectural contribution is a validated four-level intermediate representation which maps quantum programs through the hierarchy: circuit→gate→pulse→waveform.

Starting from the circuit level, the compilation workflow begins with standard OpenQASM parsing, followed by transpilation into native gate sets. This is then further compiled into calibrated pulse sequences, and then lastly waveform generation. Each stage incurs minimal latency, with total compilation times in the range of 2.8 ms to 10.2 ms, indicating that compilation overhead is negligible compared to simulation costs. Optimization techniques such as Virtual Z (VZ) gates are incorporated during compilation, reducing physical pulse operations and improving execution fidelity.

Dr Jun also mentions the inclusion of open-system modeling. The framework incorporates realistic device physics (environment coupling) through Lindblad master equation dynamics, capturing decoherence, dissipation, and leakage beyond the computational subspace. This is particularly important for superconducting transmon systems, where operations are implemented through calibrated microwave pulses rather than idealized gates.

To address scalability, the work introduces QuanTempo, a GPU-accelerated tensor network platform integrated as a backend within EmuPlat. QuanTempo supports popular Tensor Network representations such as MPS, MPO and TTN. Benchmark results demonstrate the effectiveness of this approach. In open-system simulations of the Heisenberg model under Lindblad dynamics, conventional vectorized methods exhibit exponential memory scaling (~ $O(4^N)$), restricting simulations to small system sizes (typically $N<13$). In contrast, QuanTempo achieves substantial speedups, approaching two orders of magnitude, highlighting the practical advantages of tensor network methods.

Further validation is provided through quantum-inspired simulations of the Quantum Fourier Transform. Representing QFT as an MPO ansatz enables polynomial scaling, in contrast to the exponential complexity of classical FFT applied to full quantum states. A crossover is observed approximately around 16–18, beyond which the tensor-network approach becomes significantly more efficient, with large performance gains at higher system sizes. Furthermore, Bell state simulations achieve approximately 94% fidelity under full density matrix (Uhlmann) measures and around 99.9% population fidelity, confirming that the platform accurately captures realistic device behavior while maintaining high execution accuracy.

The seminar concludes with several future directions, including integrating tensor networks with open-system dynamics in non-Markovian regimes, developing digital twin architectures for real-time hardware emulation, and incorporating AI-driven calibration methods such as reinforcement learning and transfer learning. The framework also points toward broader applications of quantum-inspired methods in optimization, quantum chemistry, and materials science.

My research focuses on AI-driven and physics-informed methods for combinatorial optimization and quantum computational problems, including variational techniques, statistical mechanics–based approaches, and hybrid quantum-classical models. The primary overlap with the speaker's work lies in the interface between algorithm design and physical execution. While my work develops optimization frameworks and energy-based models, the vQPU and EmuPlat framework operate at the level where these abstractions are realized under hardware constraints. The emphasis on pulse-level modeling, open-system dynamics, and realistic control conditions directly connects to how such algorithms would perform on actual devices, aligning my focus on algorithmic formulation with the speaker's focus on hardware-aware validation.

My research can benefit from this work by moving beyond idealized simulations toward physically grounded evaluation. Current methods are typically tested as simulatory environments. EmuPlat provides a platform to evaluate these approaches under realistic noise, decoherence, and control limitations, offering a more accurate measure of robustness. Additionally, QuanTempo's tensor network methods enable scalable analysis of larger systems, which is essential for understanding the behavior of optimization strategies beyond small-scale settings. Incorporating these tools would allow my work to evolve toward more reliable, hardware-aware models suitable for near-term quantum systems, therefore helping me develop algorithms and models that aren't just good in theory but also functional in real life scenarios.

By : Ronit Dutta [1010683] [PhD - SMT]